LATHIFAH SAHDA

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Informatics graduate specializing in NLP and LLM systems. Thesis on a constrained-LLM pipeline for news text, built on a 1,760-article dataset and evaluated on 1,440 outputs across three LLMs; research intern building an agentic verification system over a Neo4j knowledge graph (GraphRAG, MCP, LangGraph). Teaching assistant for Deep Learning and Database Systems. Open to relocation to Jabodetabek.

Education

Institut Teknologi Sepuluh Nopember (ITS) Surabaya, Indonesia
Bachelor of Computer Science, GPA 3.70/4.00 – Asrama Mahasiswa Nusantara Scholarship *Jul. 2022 – Aug. 2026*

- Thesis: hybrid pipeline (rule-based NER/temporal constraints + constrained LLM + rule-based post-processing) converting news narratives into procedural text. Built on a 1,760-article dataset; compared Llama 3.3 70B, GPT, and Claude on 480 held-out articles per model (1,440 evaluations, 16 domains): BERTScore F1 0.645–0.700, overall pass rate 98.5–100%. Post-processing raised entity preservation by 31.7–38.1 percentage points.
- **Relevant coursework:** Machine Learning, Deep Learning, Text Mining, Data Mining, Multivariate Data Analysis, Probability & Statistics, Artificial Intelligence Concepts, Knowledge-Based Systems, Database Systems.

Technical Skills

AI & NLP: LLMs, RAG / GraphRAG, AI agents (LangGraph, MCP), prompt engineering, topic modeling, PyTorch, Transformers
Data: Python, SQL, Neo4j/Cypher, data analysis, Streamlit dashboards
Software: Flask, REST APIs, Next.js, Flutter, Laravel, Git

Work Experience

Research Intern – TEEP Program Jun. 2026 – Aug. 2026
Center for IoT Innovation, National Taiwan University of Science and Technology (NTUST) *Taipei, Taiwan*

- Built a verify-then-retry agent layer over a Neo4j knowledge graph (355 nodes, 805 relations) that runs three independent gates before answering: faithfulness (LLM-as-judge), trust (confidence, source weight, recency), and temporal validity.
- Integrated the agent with GraphRAG retrieval through MCP and LangGraph. Evaluated hallucination, tool-misuse, and unsafe-action risks with agent-safety frameworks (AGENTSAFE, OpenClaw).
- Result: 95% gate accuracy (19/20) on a 20-question benchmark after manual adjudication and 85% with automated scoring alone.

Research Assistant & Laboratory Administrator Jun. 2024 – Jun. 2026
Information Intelligent Management (MCI) Laboratory, ITS *Surabaya, Indonesia*

- Research on Hybrid RAG and ReAct prompting for LLMs in education, and on dynamic topic modeling and multi-label deep-learning classification of Indonesian news.
- Co-led Kinderfin, a web-based financial management system for a kindergarten (community service): gathered requirements by survey and interview, then presented the product to the school.

Teaching Assistant 2023 – 2026
Informatics Engineering Department, ITS *Surabaya, Indonesia*

- Assisted six courses, from Deep Learning to Database Systems and Automata Theory. Guided 300+ students in lab sessions and reviewed their work.

IT Intern – Mobile Application Developer Oct. 2025 – Jan. 2026
UBS Gold *Surabaya, Indonesia*

- Built Secure Verification App, a Flutter proof of concept (Android/iOS) for approving internal HR, finance, and QMS requests, with device binding, face recognition via the company's internal API, and a WireGuard VPN.

Web Developer Intern Mar. 2025 – Jun. 2025
Directorate of Business Cooperation and Management (DKPU), ITS *Surabaya, Indonesia*

- Restructured the DKPU ITS website (WordPress). Reorganized navigation and added bilingual (ID/EN) support.

Projects & Achievements

- **Ruang Topik**, 2026 (Text Mining final project, team of 3) – scraped 20,334 articles from six national news portals and compared LDA, NMF, and BERTopic: NMF scored highest topic coherence (C_v 0.825, k=25) vs BERTopic 0.778 and LDA 0.616. Embedding choice moved coherence more than Optuna tuning (0.075 spread vs +0.021). Deployed as a Streamlit dashboard for non-technical readers.
- **Major.in** – web app recommending study majors: XGBoost model served by a Flask REST API, Next.js front-end.
- **PKM-K National Funding Recipient**, 2024 – WeapMath, a multiplayer educational math game, registered as copyright (HKI).

Leadership & Organizational Experience

General Secretary of External Affairs Feb. 2025 – Feb. 2026
HMTC ITS (Informatics Student Association) – Suar Peradaban cabinet *Surabaya, Indonesia*

- Helped coordinate a 120-member student association to deliver 20+ work programs in one term.
- Managed partnerships with companies, universities, and government agencies. Previously General Secretary of Administration II (2024).

Head of Public Relations Division (committee member since 2023) 2024
Schematics ITS (national IT event by HMTC) *Surabaya, Indonesia*

- Led the PR team's multi-channel promotion and built collaboration with 20+ media partners, helping the event reach 2,000+ participants.

Reporter & Editor Nov. 2022 – Jun. 2026
ITS News & Magazine (ITS Online), Public Communication Unit (UKP ITS) *Surabaya, Indonesia*

- Promoted from reporter to editor. Wrote and edited 150+ articles and magazine pieces on technical topics for a general audience (published portfolio on Linktree).

Other Roles & Public Speaking

- **Master of Ceremony**, IDSECCONF 2024 (Indonesia's annual IT security conference).
- **Tim Kawal Gemastik ITS**, 2025 – supported ITS teams in the national ICT competition (Gemastik).

Training & Certifications

- **TOEFL ITP**, score 520 (Listening 55, Structure 48, Reading 53), Colorado Course Bengkulu, Aug. 20, 2026
- Formulas and Functions in Microsoft Excel (online class, 2 hours; certificate of participation), AmanExcel, Oct. 31, 2026
- AI Fluency: Framework and Foundations, Claude Academy (Anthropic), course completion badge, Nov. 1, 2026
- LKMM Tingkat Menengah (Intermediate Student Management & Leadership Training), ITS, 2024
- LKMM Tingkat Dasar (Basic Student Leadership Training), Informatics ITS
- Character Building Training (ESQ), 2023